

THE MARKET-TIME RISK LAYER FOR TOKENIZED STOCKS ON X LAYER

Credit on the market's clock.

Measure the exit that is really there, and how long a loan must survive before the next deep market. Post both as signed terms on X Layer. Let anyone recompute them.

VERSION

1.1 · 25 September 2026

STANDARD

KTS 0.2 · params 2026-09-22.1

AUTHORS

Xyndicate Labs

BUILT FOR

OKX Dev Day 2026

ABSTRACT

Tokenized stocks trade around the clock; the markets behind them do not. When New York or Hong Kong is shut, no new reference price arrives and the onchain pool becomes the only exit. A lender that sizes credit against a price rather than against that exit lends more than it could liquidate at exactly the moments it would need to. Kerb measures the exit that is really there in X Layer pools, and how long a loan must survive before the next deep market, and publishes both as signed credit terms on X Layer mainnet every few minutes. Every term recomputes from its published inputs. A reference credit market, any contract, any agent and any developer read the same terms.

TERMS POSTED ON MAINNET 7,449	POOL READINGS STORED 126,465	ASSETS · MARKETS 10·4	CONTRACTS, SOURCIFY EXACT 34
-------------------------------------	------------------------------------	--------------------------	------------------------------------

Live figures read from api.usekerb.xyz at 2026-09-25 06:23 UTC. usekerb.xyz/proof checks them on every load.

01 The problem

02 Design

03 Measurement

04 The standard (KTS 0.2)

05 A worked example

06 The covenant

07 Attestation and guardrails

08 System architecture

09 Consumers

10 Evidence from the live system

11 Limits

12 What comes next

13 References

01

The problem

Three clocks run under every tokenized stock.

The exchange keeps its own calendar, with lunch breaks, holidays and early closes. Reference prices follow that calendar. Onchain liquidity follows neither: it moves with liquidity providers, incentives and the hour. Kerb's first Market-Time Report measured 35,130 readings over one closed weekend on X Layer: seven of ten asset pools lost in-range liquidity, the largest by 49%.

A collateral value that ignores this is precise and wrong. The questions a lender must answer are whether the collateral could be sold, and how long the loan must hold before it could.

02

Design

Never lend more than you can liquidate.

THE DESIGN PRINCIPLE

Every choice follows from that sentence: measure the exit rather than assume it; tie each loan's margin to the horizon it must survive; never move the liquidation line under an open loan; let anyone check every number; keep models out of anything that produces a number.

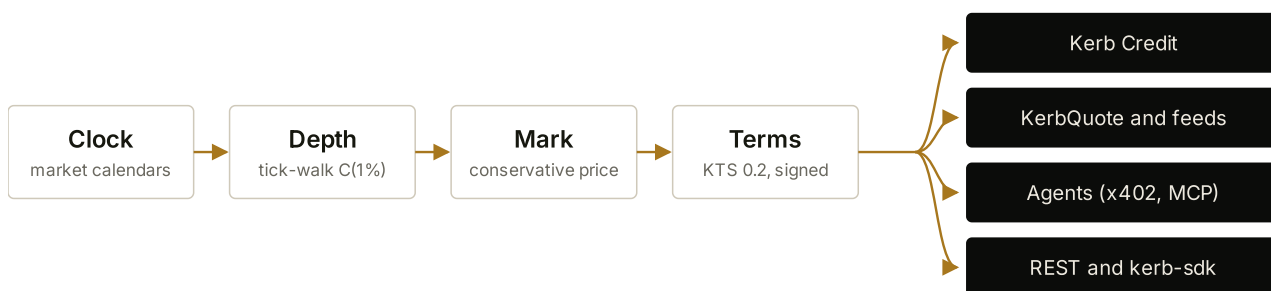


Figure 1. Four layers turn measurements into terms on X Layer mainnet; four kinds of consumer read the same terms.

03

Measurement

3.1 Clock

Each asset follows its underlying market's calendar (NYSE, Nasdaq, NYSE Arca, HKEX with its lunch break), resolved identically off chain and on chain in KerbClock; a 1,000-timestamp fuzz test holds the two in agreement. A regime machine names the state of each asset in a strict order: Halted, Stale, Corporate action, Reference closed, Pre-transition, Recovery, Thin, Normal, Deep. The first rule that holds wins.

3.2 Depth

Every minute the collector reads each pool's price, in-range liquidity and initialised ticks. The engine walks the Uniswap V3 pool tick by tick in the direction of a sale and finds $C(i)$, the largest sale that moves the price at most i . $C(1\%)$ is cross-checked against the OKX DEX aggregator at the same notionals; when the two differ by more than 25%, the smaller is used.

3.3 Mark

The Credit Mark is the lower of the reference median and the pool price along the whole path to the loan asset, less a haircut set by the regime. If the sources disagree by more than 2%, the regime becomes Stale and new borrowing stops. The Mark is conservative by construction; it is not a mid-market price.

04

The standard (KTS 0.2)

The Kerb Terms Standard turns measurements into two loan-to-value limits and two debt limits. Each mode's margin below the fixed liquidation threshold LT is tied to the horizon it must survive:

```

carryMargin      = max(floorCarry,    k · v · g(H_next_deep) + s)
sessionMargin    = max(floorSession, k · v · g(H_cure)      + s)
carryLTV         = LT - carryMargin
sessionMaxLTV    = LT - sessionMargin
debtCeiling      = kCeiling · C(1%)
maxPositionDebt  = min(capAbs, capShare · C(1%))

```

$g(H)$ is the 99th-percentile price gap over a horizon of H hours from five years of daily closes; v is a recent-volatility scaler; s is the measured exit cost at the reference liquidation size. Before a long closure, Carry tightens because its horizon grows, while Session Max, which only has to reach the next Last Call, does not.

PARAMETER	LIVE VALUE	MEANING
k	2.5	Stress multiplier on the horizon gap
floorCarry, floorSession	5%, 3%	Smallest margin each mode may keep
kCeiling	0.75	Debt ceiling as a multiple of C(1%)
capAbs, capShare	\$25,000, 25%	Largest single position
Reference liquidation size	\$10,000	Size at which exit cost s is measured
Loosen step, cooldown	2%, 30 min	How fast terms may loosen

05

A worked example

At the New York close on 21 September 2026 the horizon KOx's Carry must survive grew from 17 h 31 m to 41 h 26 m, spanning the night and the next session's gap. The stressed gap over that horizon rose from 3.63% to 5.41%.

KOX	BEFORE 20:00 UTC	AFTER	CAUSE
Carry	55.60%	51.57%	Horizon -4.11 pts, loosening cap +0.08, exit cost 0.00
Session Max	61.20%	54.47%	Margin left its floor as the cure horizon grew

The causes sum to the posted move exactly. Across 2,186 attributed changes in the record, the unexplained residual is zero. Every asset page shows its own changes with their causes.

06

The covenant

A Carry loan needs no attention. A Session Max loan borrows more now and accepts a covenant: when Last Call opens, before the market weakens, the position must come back to its Carry target. If the owner does not act, anyone may cure it, repaying only the difference for a small bonus in collateral, while liquidity is still there.

THE RULE THAT NEVER BENDS

The liquidation line never moves. Sessions move only how much may be borrowed and when a cure is due.

07

Attestation and guardrails

The attester signs each report (EIP-712) and posts it to KerbTerms with the keccak256 of its complete input bundle; every Kerb transaction carries the ERC-8021 Builder Code `kt0h16xyh1x8xmt`. KerbTerms enforces what the attester cannot change: bounded LTVs, a fixed liquidation line per asset, tightening at once and loosening in steps after a cooldown, and a maximum report age after which terms are unusable. The attester may clamp tighter than the engine, never looser.

```
# no account, no key, no database
git clone https://github.com/Franlinozz/Kerb && cd Kerb && pnpm install
pnpm --filter @kerb/engine kerb verify <inputsHash>
```

The verifier fetches the bundle, rejects any bytes that do not hash to the onchain inputsHash, recomputes every figure and compares them with the values KerbTerms holds on X Layer.

System architecture

Kerb · system architecture

One measured, deterministic term, posted on X Layer and read by a credit market, contracts, agents and code.

- 1 · SOURCES
- 2 · MEASURE AND COMPUTE (OFF CHAIN, RECOMPUTABLE)
- 3 · POST (X LAYER)
- 4 · CONSUMERS

X Layer pools
10 xStocks Uniswap V3 pools
slot0, liquidity, ticks, every minute

OKX DEX aggregator
sell quotes at fixed sizes
cross-checks the tick-walk

Reference prices
xStocks issuer data, Yahoo as a second, independent check

Exchange calendars
XNYS · XNAS · ARCX · XHKG
sessions, lunch, DST, holidays

APPS/COLLECTOR
kerb-collector → observation store
Postgres, append-only (triggers reject UPDATE and DELETE)
every row kept; bundles cut from it byte for byte

APPS/ENGINE · NO MODEL, NO FLOATS
KTS 0.2 engine (pure functions)
Clock: regime and the next weakening, per asset
Depth: C(1%) by tick-walk, the exit that is really there
Mark: the lower of reference and pool, less a haircut
Terms: Carry, Session Max, fixed LT, debt ceiling

PUBLISHED BY THE API
Input bundle
canonical JSON, hashed; anyone recomputes the same
report byte for byte (kerb verify <hash>)

APPS/ATTESTER
kerb-attester
guardrails (tighten fast, loosen slow) · EIP-712 sign
post with Builder Code Kt0h6xhix8xmt (ERC-8021)

RISK PLANE · REAL XSTOCKS
X Layer mainnet · 196
KerbTerms: signed terms + inputs hash
KerbClock: calendars and regimes
KerbQuote: max borrow, cure
deadline, usability in one call
KerbMarkFeed × 10: Credit Mark
as a Chainlink-shaped feed
Sourceify exact match, all 34

CREDIT PLANE · MIRROR TERMS
X Layer testnet · 1952
KerbCredit: Carry, Session Max,
Last Call, public cure
KerbMirror: mirror collateral
mUSDG loan asset · demo clock
kerb-demo-keeper: a standing
position to cure every cycle

Kerb Credit
borrow, Last Call, cure on testnet

Any contract
KerbQuote and Credit Mark feeds

Agents
x402 paid checks · MCP · OKX.AI

Developers
REST (no key) · npm i kerb-sdk

People
usekerb.xyz · Telegram alerts

EXPLAIN · OBSERVE · ALERT (RUN BESIDE THE CORE, NEVER IN THE POSTING PATH)
kerb-attribution
why each term moved; horizon,
volatility, exit cost, caps

kerb-indexer + API
events to Postgres, 24 REST
endpoints, free MCP server

kerb-agents
x402 on X Layer through the
OKX facilitator, USDTO

Market-Time Reports
what the market clock did to
real exit capacity, versioned

Last Call alerts
browser notifications and
@kerbAlertsBot on Telegram

Chain is canonical

Postgres is a rebuildable view

Bundles are the irreplaceable data

No model near the numbers

Unaudited · Testnet credit

----- data

----- signed post on chain

github.com/Fran1Inozz/kerb · usekerb.xyz · api.usekerb.xyz

09

Consumers

CONSUMER	HOW IT READS KERB TERMS	WHERE
Kerb Credit	Carry or Session Max, Last Call, a public cure table, a standing demo position every cycle	X Layer testnet
Contracts	KerbQuote: max borrow, cure deadline and usability in one view call. KerbMarkFeed: the Credit Mark as a Chainlink-shaped feed that fails closed	X Layer mainnet
Agents	Kerb Credit Check and Exit Check, one cent in USDT0 per call over x402 through the OKX facilitator; a free MCP server; registered on OKX.AI (agent #13887)	X Layer mainnet
Developers	A public API with no key, npm i kerb-sdk, an open hourly dataset, /llms.txt	api.usekerb.xyz

10

Evidence from the live system

Incentives ending (Report #2)

At the 07:00 UTC end of the X Liquidity campaign on 24 September, C(1%) held for all ten assets. By the 08:30 UTC capture it had fallen by 10% or more for 5 of 10, the largest HKEXCx at -82.80%. The Hong Kong close sits inside that interval, and the report says so.

The dispersion guard in production

From 24 Sep 08:13 to 25 Sep 01:43 UTC, HKEXCx's pool price sat more than 2% from its reference. The regime went to Stale and new borrowing stopped on its own; it then loosened in capped steps.

Verified, not asserted

- The full credit lifecycle ran in real browsers on the live site with fresh wallets.
- 1 paid agent call settled in USDT0 on X Layer mainnet; calls without a transaction are never counted.
- KerbQuote on mainnet returns the same max borrow as the posted terms.
- 6,342 of 6,342 exit checks follow the selection rule against OKX DEX.
- 594 TypeScript and 120 Solidity tests pass; all 34 deployments are Sourcify exact matches.

11

Limits

Kerb is unaudited. One attester signs the terms. The reference price is issuer data with an independent check, not a verified oracle report. Credit runs on testnet with mirror collateral: the

production tokens are restricted in some jurisdictions, and lending real money needs an audit first. Input bundles are served by the Kerb API since the free pinning quota ran out on 21 September; every bundle remains verifiable by its hash.

12

What comes next

1. A verified oracle reference (Chainlink Data Streams) as the first rung for the Mark.
2. Several attesters with a threshold, and an audit.
3. Curator integrations that set supply caps from measured depth.
4. Real collateral, where the law and the audit allow.

13

References

1. Kerb Terms Standard 0.1 and the 0.2 amendment: github.com/Franlinozz/Kerb/blob/main/docs/KTS-0.1.md and [docs/v2/KTS-0.2.md](https://github.com/Franlinozz/Kerb/blob/main/docs/v2/KTS-0.2.md).
2. Market-Time Reports #1 and #2: usekerb.xyz/research.
3. API reference with captured responses: github.com/Franlinozz/Kerb/blob/main/docs/API.md.
4. Contracts, posts and the live recompute: usekerb.xyz/proof. Claims and their evidence: docs/release/CLAIM_EVIDENCE.md.



Most lenders ask what collateral is worth.
Kerb asks whether you could sell it.

PRODUCT	www.usekerb.xyz
PROOF	www.usekerb.xyz/proof
SOURCE	github.com/Franlinozz/Kerb
API	api.usekerb.xyz
SDK	<code>npm i kerb-sdk</code>
STUDIO	Xyndicate Labs · @xyndicatepro

Kerb is unaudited software. Kerb Credit runs on X Layer testnet with mirror collateral that has no claim on any security. Nothing in this paper is investment advice. See usekerb.xyz/legal/risk.